Edge-girth-regular graphs: complete generation, numerical bounds, and the quartic one-hexagon obstruction

Human Compression project

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Abstract

We assemble proofs and finite certificates for the principal mathematical results in the June 2024 edition of Goedgebeur and Jooken's *Exhaustive generation of edge-girth-regular graphs*. The synthesis contains a corrected complete-generation theorem, all 142 lower-bound rows, 84 finite upper witnesses, 47 asserted exact counts, and 118 minima in two symmetry classes. The printed first-phase generation rule and one symmetry minimum require explicit corrections. A computer-assisted local argument excludes quartic graphs on 54 vertices having exactly one hexagon through each edge; together with the smaller-order argument this gives $57 \leq n(4,6,1) \leq 84$. The companion proofs state the external classification and classical algebraic premises used in their respective rows. Neither finite-table completeness nor this new local argument settles the three remaining assertions of the source's Conjecture 2.

1 Definitions, scope and conclusions

All graphs are finite, simple and connected unless a partial graph is explicitly being considered. An $\mathrm{egr}(n,k,g,\lambda)$ graph has n vertices, is k-regular, has girth g, and has exactly $\lambda > 0$ unoriented cycles of length g through every edge. Let $n(k,g,\lambda)$ be its least possible order, with value ∞ if no such graph exists. Write n_{vt} and n_{at} for minima restricted to vertex-transitive and arc-transitive graphs, respectively. These restricted minima need not be unrestricted minima.

The fixed source throughout is arXiv:2401.08271v2, dated 25 June 2024 [1]. Its physical pages 17–20 contain Tables 1–4; physical page 21 contains Table 5. The appendices below retain each numerical claim separately from any stronger conclusion of its companion proof. An infinity symbol in the source's upper-bound column can mean that no finite example was reported. It is not, by itself, a nonexistence theorem.

Theorem 1 (Numerical synthesis)**.** *Subject to the external premises explicitly specified by the cited companion proofs, every lower inequality and every finite upper inequality in Appendix A holds. Its 47 displayed exact counts are counts of isomorphism classes at the stated upper order. The 118 corrected symmetry minima in Appendix B hold in their stated symmetry classes.*

Here "exact" applies only where the source asserts an exact count. A supplied graph for a row marked "at least one" is not an exhaustion of that row's extremal graphs. In particular, the last quartic row proved here is a pair of inequalities, not an equality or an exact count.

Proof. The lower and exact-count columns refer to the numbered companion proofs in Appendix D. Each of the 142 rows is joined to its lower-bound proof; each of the 47 exact counts is separately joined to its exhaustion or structural classification. The common positive-witness proof C1 checks the 84 finite upper inequalities. The symmetry proof C4 supplies the separate catalogue exhaustion and symmetry certificates for the 118 rows. The only previously missing lower inequality, for $(k,g,\lambda) = (4,6,1)$, follows from Theorem 5 below. The machine-readable

concordance preserves the literal source rows and identifies the exact companion editions; the mathematical statements of those companions, rather than a numerical agreement or a solver status, supply these implications. □

Among the source's highlighted conclusions are

$$n(3,6,2) = 24, \qquad n(3,8,8) = 40, \qquad n(3,9,6) = n(3,9,8) = 60,$$
$$n(4,5,1) = 30, \qquad n(4,6,9) = 35, \qquad n(6,5,20) = 42.$$

The first two use the cubic catalogue argument C9; the two order-60 statements use the unrestricted girth-nine argument C3, including its explicit cage-classification premise. They do not follow from a vertex-transitive minimum alone. The remaining three use C6, C30 together with the lower and positive proofs, and C2 with C1, respectively. There are two isomorphism classes at $(24, 3, 6, 2)$ and one at $(40, 3, 8, 8)$, $(30, 4, 5, 1)$ and $(35, 4, 6, 9)$. No exact count at order 60 or for the displayed order-42 row is asserted here.

2 Cycle counting and complete generation

We state the mathematical interface of the complete-generation proof [2]. It is independent of the historical running times or the adequacy of fixed arrays in an old implementation.

Lemma 2 (Deleted-edge counting)**.** *Let H have girth at least g and let uv be an edge. In $H - uv$, perform breadth-first search from u. Set $a(u) = 1$ and*

$$a(x) = \sum_{y \sim x,\ d(u,y) = d(u,x) - 1} a(y).$$

If $d(u, v) = g - 1$, the number of g-cycles through uv is $a(v)$; otherwise it is zero. The computation takes $O(|V(H)| + |E(H)|)$ arithmetic operations.

Proof. Deleting uv identifies its cycles with simple u, v paths. Girth forbids a path shorter than $g - 1$, so a path of that length is shortest. Every shortest path has a unique penultimate vertex in the preceding layer, giving the recurrence. Before the FIFO queue removes a vertex, all preceding-layer contributions have arrived. Each adjacency is examined at most twice. □

For comparisons with λ, counts can be capped at $\lambda + 1$. The identity $\min(C, a + b) = \min(C, \min(C, a) + \min(C, b))$ proves that this preserves the decisions below, equal to, or above λ. Arithmetic complexity is not an assumption of constant-time unbounded integers; the capped additions use $O(\log(\lambda + 2))$ bits.

For odd girth $2s + 1$, start with the degree-k tree of depth s about a vertex, grouping leaves by their first root edge. For even girth $2s$, start with an edge and the two degree-k trees of depth $s - 1$ on its sides. Add isolates to reach the target order. Girth prevents collisions inside these trees. A horizontal edge joins different leaf groups. The tree path it closes is a girth cycle, and conversely every girth cycle through a prescribed root edge (or the central edge in the even case) arises this way. Consequently each group has horizontal incidence quota λ.

Theorem 3 (Corrected generation)**.** *The following serial depth-first scheme covers every target up to isomorphism and, with exact terminal keys, outputs it once. Add only edges between deficient vertices that preserve girth and do not force more than λ girth cycles through an edge. In the first phase choose a* deficient leaf group *and attempt all eligible horizontal edges incident with that group. Once all quotas are met, choose an unsaturated nonisolated vertex and attempt all eligible incident edges. The scheme may use exact uncoloured keys on the entire graph, literal rollback, historical sibling exclusions, and the least-labelled isolated-neighbour representative as specified in [2]. The final test requires connectedness, degree k and cycle count λ at every edge.*

Proof. Every target contains a seed. If a group is deficient, some missing target edge is horizontal and incident with that group. In the second phase an unsaturated nonisolated vertex likewise has a missing target edge. If every nonisolated vertex is saturated while isolates remain, there is no connected completion. Degree-capacity and cycle-excess tests are necessary restrictions. Girth-cycle counts cannot decrease when edges are added.

The less immediate pruning argument is chronological. A recursive call strictly increases edge count, so a previous state with the same exact uncoloured key cannot be an ancestor: its call has already finished. Earlier sibling edges are excluded only after their entire child calls finish. Inductively, a target completion removed by such a historical exclusion is already represented by an earlier completed call. If no target edge is historically excluded, the deficient-group or selected- vertex argument supplies an eligible target edge. Relabelling isolated vertices justifies retaining the least isolated neighbour; when its representative edge is historically excluded, the same chronological argument supplies earlier coverage. It is not necessary to assert that the historical exclusion masks are isomorphic.

The complete induction, including capacity pruning in the presence of these masks, is given in [2]. Restoration returns exactly the parent graph and records before a new sibling exclusion is installed. There are at most $nk/2$ edge additions on a branch. The exact terminal test proves soundness, and the terminal isomorphism keys give uniqueness. □

The proof also admits the one-phase version and delayed excess-cycle tests, with the same exact terminal predicate. A bounded memo table may omit storing a nonterminal state; doing so loses a pruning opportunity, not a completion. Duplicate-free output still requires the stated terminal-key rule.

2.1 The printed first-phase rule

Printed Algorithm 2 chooses a minimum-option leaf, which can lie in a group whose quota is already full. For $(n, k, g, \lambda) = (9, 4, 3, 1)$, take the star with centre 0, leaves $1, 2, 3, 4$, and four isolates. After adding 12, leaves 1 and 2 each have four valid neighbours, all isolated; leaves 3 and 4 have five, including each other. The printed rule chooses 1 or 2 but then has no horizontal edge to attempt. It misses 34 and therefore misses the 3×3 rook graph, which is connected, 4-regular and has exactly one triangle through each edge. The separate literal counterexample checks all tie and first-edge histories.

The correction is to loop over a deficient group, as in Theorem 3. The later December 2024 C source, at commit `41022f4a3ad84e4502f8889c5b64132225411ca2`, already uses this corrected loop. The counterexample to the printed June rule is not a counterexample to that later implementation. The abstract theorem and exact reference implementation do not retrospectively certify every historical machine execution; the numerical claims have their own proofs and certificates.

3 Quartic graphs with one hexagon through every edge

Let Q denote the hypotheses of a connected quartic graph of girth six with exactly one hexagon through each edge. The following argument has no bipartiteness assumption.

3.1 The smaller-order reduction

The two hexagons through a vertex pair its four incident edges. Distinct hexagons meet in at most one vertex. The distance spheres of radii $0, 1, 2, 3$ about a vertex have sizes

$$1, \quad 4, \quad 12, \quad 34.$$

Indeed there are 36 length-three nonbacktracking paths starting at the vertex; precisely the two antipodal pairs on its two hexagons identify their ends. A further collision would force an edge

into two hexagons. Thus $|V(G)| \geq 51$. Counting edge–hexagon incidences gives $6h = 2|V(G)|$, so 3 divides $|V(G)|$.

We import the complete order-51 spectral lemma [3]. At equality the radius-three ball is the whole graph, and the adjacency matrix A and the antipodal two-factor matrix M satisfy

$$M = A^3 + A^2 - 6A - 3I - J.$$

The proof compares the rational multiplicities of the cycle eigenvalues of M with the cubic preimages under this polynomial. Its cyclotomic integer-ring premise and finite residue certificates are explicit in [3]. The resulting multiplicity congruences contradict the principal eigenspace. This excludes order 51; we do not replace that lemma with a numerical spectral approximation.

3.2 A complete local cover at order 54

For a hexagon C of a hypothetical order-54 graph, write $t(C)$ for the number of vertices at distance at least three from C. The local incidence lemmas [4] give twelve first-layer vertices. The second layer consists of vertices S with one first-layer parent and vertices T with two parents. The remaining vertices form Z, with

$$|T| = |Z| = t(C), \qquad |S| = 36 - 2t(C).$$

The previously proved three- and four-exterior exclusions, together with the layer restrictions, leave $t(C) \in \{0, 5, 6\}$.

For $t = 5$, the auxiliary graph on Z has the three coloured hosts and actual parent assignments of the complete five-exterior theorem. Its three necessary formulas have separate reconstructed refutations. Thus $t = 5$ is impossible. This uses the full five-exterior cover; no four-exterior filter is transferred to it without a proof.

For $t = 6$, each T gives an edge between its two Z neighbours. The resulting simple graph R has six vertices and six edges, with colours recording the opposite anchors on C. The parent-cover theorem excludes the fork and ear configurations and gives eight uncoloured shapes, twelve coloured rows and seventy raw parent records. The letters N, L, R name actual parent choices. Disconnected hosts, a leaf, or an isolated Z vertex are allowed where the theorem permits them.

Each raw record gives a host with 54 vertices, 66 fixed edges and the complete allowed optional SS, SZ and ZZ pairs. The fixed edges are the six cycle edges, twelve edges to the first layer, twelve parent edges to T, twenty-four parent edges to S, and twelve TZ edges. Shared parents change the individual deficits, not these totals. Optional pairs include the isolated-Z possibilities; their host degree can be 29. No degree-24 assumption from the five-exterior case is used.

Literal permutations of all 54 vertices take these seventy domains to the twenty-four domains in Appendix C. Both the entire fixed-edge set and the entire optional-edge set are checked under each permutation. These equalities transport every graph completion and its cycle properties. A claim that the representatives are mutually nonisomorphic is unnecessary.

3.3 Necessary formulas and their refutations

There is one Boolean variable for each optional edge. The original completion formula imposes degree four and the necessary girth restrictions. Every one of the twelve fixed TZ edges must occur in a hexagon. Enumerating simple length-five paths between its ends in the literal finite host gives the complete candidate list. Cycles are discarded only for the proved degree or shortcut obstructions. A shared positive gate is equivalent to the conjunction of the optional edges of a cycle; the disjunction of these gates is required for each fixed TZ edge. The catalogue is therefore complete even when a host has an isolated Z vertex.

A short cycle, or two distinct hexagons sharing an edge, gives a necessary negative clause on its optional edges. Every universal clause and every subsequent lazy cut carries such a literal

witness. Hence a Q completion would satisfy the entire final formula, after extending its edge assignment to the gate variables. Only this necessary direction is used.

The independent saved-certificate procedure recomputes the selected catalogue, validates every cut witness, rebuilds the complete DIMACS formula in its fixed clause order, and compares it with the formula actually refuted. An unchanged reverse-unit-propagation checker then validates all proof additions and an explicit empty clause. Its active and historical clauses are consequences of that same original formula; the solver's own UNSAT declaration is not a premise. Dimensions and checked additions for all twenty-four domains appear in Appendix C.

One domain's accepted proof is a separate copy of its original trace followed by an empty-clause line. This is not an unconditional rule for repairing a trace: the unchanged checker accepted the resulting complete trace against the reconstructed original formula. A previous failed read remains failed. Another domain uses a separately completed proof of its already fixed formula. Neither case repeats or assumes the correctness of an incomplete search.

It follows that no hexagon of an order-54 Q graph can have $t = 6$: its completion would transport to one of the refuted domains and induce a satisfying assignment of that domain's necessary formula.

3.4 The global contradiction and lower bound

Theorem 4. *There is no order-54 graph satisfying Q.*

Proof. There are 108 edges and therefore 18 hexagons. Their incidence graph is simple and 6-regular: the two hexagons through each vertex are distinct, and two hexagons meet at most once. The zero-neighbour lemma [4] proves that two zero hexagons cannot meet a zero hexagon at opposite vertices. Thus a zero hexagon has at most three zero neighbours and at least three positive neighbours, where positive means $t(C) > 0$.

If n_0, n_+ count zero and positive hexagons, then

$$3n_0 \le 6n_+, \qquad n_0 + n_+ = 18,$$

so $n_+ \ge 6$. But every hexagon has $t \in \{0, 5, 6\}$, and the complete local refutations exclude both positive values. This is a contradiction. The argument excludes the all-zero case; it does not assert that a single zero hexagon is impossible independently of the others. □

Theorem 5. *One has* $57 \le n(4, 6, 1) \le 84$.

Proof. The sphere bound gives order at least 51 and the incidence count gives divisibility by three. The order-51 lemma and Theorem 4 exclude the only possible orders below 57. The explicit 84-vertex graph `UntilOrder250_427` satisfies the defining properties by the independent positive-witness tests C1. Thus the upper inequality holds. □

This proves exactly the lower/upper claim in source Table 2. It neither excludes order 57 nor determines the exact minimum or number of graphs at order 84.

The source's even-girth nonbipartite example also follows from the separately classified $\mathrm{egr}(15, 6, 4, 16)$ graph. In any nonempty regular bipartite graph, counting edges from the two parts gives equal part sizes. Its order is therefore even. The accepted order-15 witness is nonbipartite, so the proposed bipartiteness analogue for minimum edge-girth-regular graphs fails, as observed by the source authors.

4 Literal certificates for the illustrated graphs

The following certificates complete the descriptive claims in the source's Figures 2 and 4. They use the same three graphs as the positive witness collection C1: `UntilOrder250_50`, `UntilOrder250_82` and `UntilOrder250_300`, of orders 16, 20 and 40, respectively. All labels

below are zero-based labels of those exact graph6 records. An array p lists $p(0), p(1), \ldots$; line breaks within an array do not restart its indices. Complete arrays and graph6 strings are also retained in the literal figure certificate.

4.1 The Moebius–Kantor graph and its extension

Define $G(8,3)$ on u_i, v_i, with i taken modulo eight, by the edges u_iu_{i+1}, u_iv_i and v_iv_{i+3}. Label u_i by i and v_i by $8+i$. The following full adjacency isomorphism identifies the source's order-16 graph with this Moebius–Kantor graph:

$$p = \big[0, 1, 7, 8, 4, 3, 12, 5, 10, 14, 2, 9, 6, 11, 13, 15\big].$$

The permutations $u_i \mapsto u_{i+1}$, $v_i \mapsto v_{i+1}$ and $u_i \mapsto v_{3i}$, $v_i \mapsto u_{3i}$ preserve all three edge types and act transitively together. Transporting them to the source labels gives the two checked automorphisms

$$a_1 = \big[1, 10, 0, 11, 7, 4, 14, 12, 13, 15, 5, 8, 2, 6, 9, 3\big].$$
$$a_2 = \big[3, 13, 14, 0, 6, 11, 4, 15, 12, 10, 9, 5, 8, 1, 2, 7\big].$$

Their orbit of vertex zero contains all sixteen vertices. The following injection identifies an induced copy of this graph in the source's order-20 graph:

$$\iota = \big[1, 8, 17, 16, 2, 19, 18, 11, 3, 4, 9, 10, 14, 12, 15, 13\big].$$

The complementary set is $D = \{0, 5, 6, 7\}$. It is independent; every vertex outside D has exactly one neighbour in D, and each vertex of D has four neighbours outside it. Checking all induced adjacencies and these attachments verifies the whole add-four-vertices construction in Figure 2. It also proves connectedness, since the induced $G(8,3)$ is connected and each added vertex meets it.

Let A be the adjacency matrix of this order-20 graph. Exact integer multiplication gives

$$(A^6)_{vv} = \begin{cases} 292, & v \in D, \\ 298, & v \notin D. \end{cases}$$

This diagonal counts closed walks of length six and is invariant under every automorphism. The two values therefore exclude vertex transitivity. The literal cycle $(0, 1, 8, 9, 3)$ is a pentagon, so the graph is not bipartite. To exclude edge transitivity, suppose a connected graph is edge-transitive. The orbits of the two endpoints of one edge cover all vertices. If those orbits coincide the graph is vertex-transitive. If they differ, every edge joins the two orbits, giving a bipartition. Both alternatives fail here. Thus the order-20 graph is neither vertex-transitive nor edge-transitive.

4.2 The order-40 graph

For the source's order-40 graph, the following three permutations are checked on the entire adjacency relation:

$$b_1 = \big[\begin{matrix} 0, 1, 3, 2, 5, 4, 7, 6, 8, 9, 11, 10, 12, 15, 14, 13, 17, 16, 21, 20, \\ 19, 18, 23, 22, 33, 31, 30, 32, 29, 28, 26, 25, 27, 24, 34, 38, 39, 37, 35, 36 \end{matrix}\big].$$

$$b_2 = \big[\begin{matrix} 0, 2, 3, 1, 5, 4, 8, 6, 7, 9, 12, 13, 11, 14, 15, 10, 17, 18, 19, 20, \\ 21, 16, 24, 25, 26, 27, 28, 29, 32, 30, 33, 22, 31, 23, 35, 36, 37, 39, 34, 38 \end{matrix}\big].$$

$$b_3 = \big[\begin{matrix} 1, 34, 37, 0, 18, 21, 9, 12, 14, 8, 6, 33, 32, 7, 27, 24, 30, 5, 31, 26, \\ 4, 25, 10, 16, 20, 15, 38, 36, 11, 17, 13, 35, 19, 39, 22, 29, 3, 23, 2, 28 \end{matrix}\big].$$

Their generated orbit of vertex zero has forty vertices. This proves the vertex-transitivity assertion of Figure 4. These are positive automorphism and orbit certificates; no assertion that the displayed generators generate the full automorphism group is needed.

All finite assertions in this section were independently checked from the complete literal arrays, including the full closed-walk diagonal. The separate companion arguments still supply the defining graph parameters, uniqueness and extremality claims. The present certificates do not infer those conclusions from symmetry or an embedding.

5 External premises, corrections and attribution

The companion proofs distinguish a direct certificate from a borrowed classification. In particular, C9 uses the specified complete cubic catalogues, C3 uses the classified cubic girth-nine cages, and C4 uses the Potočnik–Spiga–Verret catalogues of cubic vertex-transitive graphs through order 1280 and tetravalent arc-transitive graphs through order 640. Checking every retained record does not prove those external catalogues complete. Positive symmetry certificates prove transitivity by explicit automorphisms and orbits, without requiring a full automorphism-group computation.

The elementary lower-bound and seed arguments retain the source's attribution to Drglin, Filipovski, Jajcay and Raiman. The order-51 proof retains its stated theorem about cyclotomic integer rings. Other borrowed inputs and their exact applicability remain in the corresponding companion manuscript; combining the rows does not discharge those premises. The proof-source concordance includes each such manuscript rather than replacing it with a claim of wholly independent rederivation.

There are two further corrections beyond the printed generation rule. First, C4 proves

$$n_{\mathrm{vt}}(3, 14, 56) = 486,$$

correcting 506 in Table 5. The literal catalogue graph has 729 edges and 2916 fourteen-cycles, so $2916 \cdot 14 = 729 \cdot 56$; independent edgewise tests and explicit vertex-transitivity witnesses supply its positive certificate. The complete catalogue comparison excludes smaller orders. The other 117 minima agree with the fixed source. Second, this graph contradicts the unqualified physical-page-12 sentence denying a cubic connected vertex-transitive graph on 486 vertices. We do not silently insert a girth restriction into that sentence.

The source authors' refutations of the earlier extremality conjecture are preserved as their results: the order-40 witness at $(3, 8, 8)$ is smaller than 54, and the order-162 witness at $(3, 12, 32)$ is smaller than 486. This synthesis makes no new priority claim for either.

The source's separate Conjecture 2 asserts four infinite minima. The previously proved companion C34 establishes $n(3, 8, 14) = \infty$ by a local forced-configuration and protected-edge argument independent of the ambient order. This strengthens the source's finite lower bound 64 and resolves its fourth clause within the acquired component collection. For the other three clauses this synthesis supplies only

$$n(3, 7, 6) \geq 42, \qquad n(3, 8, 10) \geq 48, \qquad n(3, 8, 12) \geq 48.$$

It does not prove their proposed infinity values or assess their current status elsewhere. This corrects the overbroad wording in an earlier internal completion plan that described all four clauses as unresolved. No new conjecture credit arises from assembling the already proved C34.

6 Reproduction boundary

The mathematical payload consists of the companion proofs, exact input graphs and classifications with their stated provenance, local-domain models and vertex permutations, formula construction and witness-checking sources, and complete accepted refutations. A source-only

rebuild of the RUP checker is sufficient; a historical platform-specific executable is not a mathematical premise. Formula construction, certificate verification and finite coverage are distinct from rerunning certificate discovery.

All 49 component defaults and the five overarching route families have separately accepted private executions. The overarching families cover the 29 complete saved refutations, the seventy-to-twenty-four domain transport, the literal figure certificates, the order-51 modular certificates, and the two algorithm defaults. The current concordance joins the 16 principal claim objects, all 260 source-table rows and all 53 direct written-source or composition records to those exact gates. These counts do not remove the external premises stated above or grant new conjecture or historical-running-time claims.

The staged scientific interfaces preserve the complete original defaults, their controls and their full report comparisons. Their current editions include the separately accepted catalogue-inventory and matching-manifest repairs. The package-relative binding is at `scripts/edge_girth_regular_graphs/execution/execute.py`; its immutable plan declares 81 bounded application configurations. It binds exact acquired worker sources to package paths and allows only explicitly named package or environment fields to relocate. The literal figure checker remains a separate relative entrypoint with its acquired isolation boundary. The readable reproduction guide lists these routes and distinguishes metadata preparation from actual scientific execution.

The publication closure retains the selected mathematical and runtime payloads, all written-source joins, current adapters, mode and runtime manifests, and full case expectations. Historical fixed-host supervisors, an entire old receipt roster and old machine binaries are not prerequisites merely because a prior gate records their identities. Records actually read as scientific comparisons or written premises are retained. Documentary identities remain explicit and are not silently treated as newly executed files. No old artifact or Git history is removed by this publication mechanism.

The first bounded binding targets the acquired macOS/Darwin ownership mechanism, including its `libproc` process-group observations; it is not a cross-platform process-reaping claim. Python and its required packages, the compiler and sanitizer runtime, and the specified nauty headers and library are explicit external environment premises. Canonical and classification conventions remain those of the companion proofs. The typed package-relative binding has a distinct source review. Its metadata generator derives file identities and path projections from logical roles, without requiring the author's old host paths as configuration keys. The actual focused metadata controls checked all 81 case inverses and 20 rejections. A separately owned complete dense-bounds integration reproduced its unchanged original report through this interface. These observations cover the binding boundary and one scientific route in the recorded environment. The publication admission at `data/edge_girth_regular_graphs/admission.json` must separately bind the exact assembly, manuscript, rendered PDF, registration and scoped selected-checkout evidence. The accepted mathematical executions do not by themselves assert that an arbitrary fresh machine can run every route. The reproduction guide gives the concrete prerequisites and commands; its instructions do not substitute for those actual admission records.

A Bounds and asserted exact counts

L_s is the fixed source's lower bound, L the proved lower bound, and U_s its finite upper bound. A dash in U_s means no finite upper claim, not a proof of nonexistence. The column $c(U_s)$ gives only asserted exact isomorphism counts; its dash leaves that count unspecified. C_L, C_c identify the lower and counting companion proofs below. All finite upper witnesses use C1. An infinite L is an all-order nonexistence conclusion of its cited component.

Source Table 1

k	g	λ	L_s	L	U_s	$c(U_s)$	C_L	C_c
3	3	2	4	4	4	1	2	7
3	4	2	8	8	8	1	2	7
3	4	4	6	6	6	1	2	7
3	5	2	20	20	20	1	2	9
3	5	4	10	10	10	1	2	7
3	6	2	24	24	24	2	9	9
3	6	4	18	18	18	1	2	9
3	6	6	16	16	16	1	2	9
3	6	8	14	14	14	1	2	9
3	7	2	42	42	56	—	9	—
3	7	4	28	28	28	1	2	9
3	7	6	42	42	—	—	9	—
3	7	8	∞	∞	—	—	2	—
3	8	2	56	56	64	—	49	—
3	8	4	48	48	48	—	9	—
3	8	6	48	48	48	—	9	—
3	8	8	40	40	40	1	9	9
3	8	10	48	48	—	—	9	—
3	8	12	48	48	—	—	9	—
3	8	14	64	∞	—	—	34	—
3	8	16	30	30	30	1	2	9

Source Table 2

k	g	λ	L_s	L	U_s	$c(U_s)$	C_L	C_c
4	3	1	9	9	9	1	2	7
4	3	2	6	6	6	1	2	7
4	3	3	5	5	5	1	2	7
4	4	1	18	18	18	4	5	5
4	4	2	13	13	13	1	2	7
4	4	3	14	14	14	1	7	12
4	4	4	21	21	—	—	23	—
4	4	5	10	10	10	1	2	7
4	4	6	10	10	10	1	2	7
4	4	7	∞	∞	—	—	2	—
4	4	8	∞	∞	—	—	2	—
4	4	9	8	8	8	1	2	7
4	5	1	30	30	30	1	6	6
4	5	2	30	30	30	—	46	—
4	5	3	30	30	55	—	46	—
4	5	4	30	30	—	—	46	—
4	5	5	24	24	24	1	45	45
4	5	6	20	20	20	1	2	26
4	5	7	35	∞	—	—	33	—
4	5	8	65	∞	—	—	32	—
4	5	9	∞	∞	—	—	2	—
4	6	1	57	57	84	—	50	—
4	6	2	51	51	96	—	31	—
4	6	3	45	47	90	—	31	—
4	6	4	45	45	60	—	31	—
4	6	5	42	45	81	—	31	—
4	6	6	40	41	64	—	31	—
4	6	7	39	39	60	—	31	—
4	6	8	39	39	48	—	31	—
4	6	9	35	35	35	1	2	30

k	g	λ	L_s	L	U_s	$c(U_s)$	C_L	C_c
4	6	10	39	39	48	—	44	—
4	6	11	36	36	42	—	2	—
4	6	12	35	35	40	—	43	—
4	6	13	36	36	60	—	41	—
4	6	14	36	36	—	—	42	—
4	6	15	33	33	40	—	40	—
4	6	16	33	33	36	—	2	—
4	6	17	33	33	—	—	2	—
4	6	18	32	32	32	1	39	39
4	6	19	36	36	—	—	38	—
4	6	20	36	36	—	—	37	—
4	6	21	30	30	30	1	25	29
4	6	22	39	∞	—	—	36	—
4	6	23	42	∞	—	—	35	—
4	6	24	28	28	28	1	2	24
4	6	25	∞	∞	—	—	2	—
4	6	26	∞	∞	—	—	2	—
4	6	27	26	26	26	1	2	11

Source Table 3

k	g	λ	L_s	L	U_s	$c(U_s)$	C_L	C_c
5	3	2	12	12	12	1	2	7
5	3	4	6	6	6	1	2	7
5	4	2	24	24	32	—	2	—
5	4	4	16	16	16	1	2	13
5	4	6	20	20	—	—	22	—
5	4	8	20	20	—	—	22	—
5	4	10	20	20	—	—	15	—
5	4	12	12	12	12	1	2	7
5	4	14	∞	∞	—	—	2	—
5	4	16	10	10	10	1	2	7
5	5	2	44	44	60	—	48	—
5	5	4	38	38	66	—	2	—
5	5	6	36	36	—	—	2	—
5	5	8	34	34	36	—	2	—
5	5	10	32	32	—	—	2	—
5	5	12	32	32	32	—	2	—
5	5	14	42	48	—	—	47	—
5	5	16	∞	∞	—	—	2	—

Source Table 4

k	g	λ	L_s	L	U_s	$c(U_s)$	C_L	C_c
6	3	1	15	15	15	1	2	10
6	3	2	12	12	12	1	2	7
6	3	3	9	9	9	1	2	7
6	3	4	8	8	8	1	2	7
6	3	5	7	7	7	1	2	7
6	4	1	32	36	40	—	2	—
6	4	2	30	32	36	—	2	—
6	4	3	28	28	32	—	2	—
6	4	4	25	25	27	—	2	—

k	g	λ	L_s	L	U_s	$c(U_s)$	C_L	C_c
6	4	5	24	24	24	—	2	—
6	4	6	24	24	24	—	21	—
6	4	7	24	24	36	—	19	—
6	4	8	22	22	26	—	20	—
6	4	9	20	20	20	2	2	19
6	4	10	20	20	20	5	17	19
6	4	11	20	20	—	—	2	—
6	4	12	20	20	20	—	18	—
6	4	13	20	20	20	—	16	—
6	4	14	18	18	18	1	16	17
6	4	15	20	20	—	—	16	—
6	4	16	15	15	15	1	2	8
6	4	17	16	16	16	1	2	14
6	4	18	22	22	—	—	15	—
6	4	19	24	24	—	—	15	—
6	4	20	14	14	14	1	2	8
6	4	21	∞	∞	—	—	2	—
6	4	22	∞	∞	—	—	2	—
6	4	23	∞	∞	—	—	2	—
6	4	24	∞	∞	—	—	2	—
6	4	25	12	12	12	1	2	7
6	5	1	65	65	—	—	2	—
6	5	2	60	60	—	—	2	—
6	5	3	60	60	—	—	2	—
6	5	4	60	60	—	—	2	—
6	5	5	57	57	—	—	2	—
6	5	6	60	60	—	—	2	—
6	5	7	55	55	—	—	2	—
6	5	8	55	55	—	—	2	—
6	5	9	55	55	—	—	2	—
6	5	10	52	52	—	—	2	—
6	5	11	55	55	—	—	2	—
6	5	12	50	50	—	—	2	—
6	5	13	50	50	—	—	2	—
6	5	14	50	50	—	—	2	—
6	5	15	47	47	57	—	2	—
6	5	16	50	50	—	—	2	—
6	5	17	45	45	—	—	2	—
6	5	18	45	45	—	—	2	—
6	5	19	45	45	—	—	2	—
6	5	20	42	42	42	—	2	—
6	5	21	45	45	—	—	2	—
6	5	22	40	40	40	1	2	27
6	5	23	45	45	—	—	28	—
6	5	24	45	45	—	—	28	—
6	5	25	∞	∞	—	—	2	—

B Corrected symmetry minima

For $k = 3$ the class is vertex-transitive, and for $k = 4$ it is arc-transitive. Both the printed value and the actual minimum are shown. The complete comparison is C4.

k	g	λ	Source	Minimum
3	9	2	108	108
3	9	4	408	408

k	g	λ	Source	Minimum
3	9	6	60	60
3	9	8	60	60
3	10	2	160	160
3	10	4	120	120
3	10	10	112	112
3	10	12	110	110
3	10	16	90	90
3	10	20	80	80
3	11	2	1012	1012
3	11	4	1012	1012
3	11	8	506	506
3	12	2	384	384
3	12	4	384	384
3	12	6	512	512
3	12	8	272	272
3	12	10	240	240
3	12	12	256	256
3	12	14	256	256
3	12	16	234	234
3	12	18	256	256
3	12	20	234	234
3	12	22	216	216
3	12	24	204	204
3	12	26	192	192
3	12	28	182	182
3	12	32	162	162
3	12	34	168	168
3	12	36	162	162
3	13	26	384	384
3	14	2	1092	1092
3	14	6	1092	1092
3	14	10	1008	1008
3	14	14	768	768
3	14	16	1008	1008
3	14	28	504	504
3	14	42	512	512
3	14	44	406	406
3	14	56	506	486
3	15	10	1248	1248
3	15	12	1280	1280
3	15	20	864	864
3	15	30	816	816
3	15	32	620	620
3	16	32	1250	1250
3	16	72	1240	1240
3	16	80	1012	1012
3	16	96	1008	1008
3	16	104	1280	1280
4	7	1	252	252
4	7	2	224	224
4	7	3	273	273
4	7	4	224	224
4	7	7	160	160
4	7	9	91	91
4	7	14	80	80
4	8	1	612	612
4	8	2	320	320

k	g	λ	Source	Minimum
4	8	3	336	336
4	8	4	252	252
4	8	5	384	384
4	8	6	288	288
4	8	7	512	512
4	8	8	243	243
4	8	9	320	320
4	8	10	240	240
4	8	11	336	336
4	8	12	288	288
4	8	13	256	256
4	8	14	256	256
4	8	15	204	204
4	8	16	240	240
4	8	17	224	224
4	8	18	256	256
4	8	19	240	240
4	8	20	162	162
4	8	21	192	192
4	8	22	256	256
4	8	23	256	256
4	8	24	126	126
4	8	25	192	192
4	8	26	160	160
4	8	27	168	168
4	8	28	144	144
4	8	29	192	192
4	8	30	160	160
4	8	33	160	160
4	8	36	140	140
4	8	37	128	128
4	8	39	128	128
4	8	40	135	135
4	8	41	128	128
4	8	43	128	128
4	8	45	128	128
4	8	48	110	110
4	8	60	100	100
4	8	65	96	96
4	9	6	504	504
4	9	9	600	600
4	9	15	546	546
4	9	18	506	506
4	9	19	504	504
4	9	21	285	285
4	9	27	320	320
4	9	36	320	320
4	10	50	625	625
4	10	60	546	546
4	10	61	640	640
4	10	65	640	640
4	10	67	640	640
4	10	70	624	624
4	10	71	640	640
4	10	80	546	546
4	10	90	512	512
4	10	95	576	576

k	g	λ	Source	Minimum
4	10	108	420	420
4	10	110	432	432

C The twenty-four local refutations

"Raw domains" counts the members of the complete seventy-record parent cover transported to that representative. It does not count graph isomorphism classes. Every row has a checked empty-clause conclusion against its reconstructed necessary formula.

ID	Shape and parents	Raw domains	Variables	Clauses	RUP additions
0	`c6_opposite/NNN`	1	15261	85921	445920
1	`c6_opposite/NNL`	6	15861	83147	43465
4	`c6_opposite/NLL`	6	16231	81194	21594
5	`c6_opposite/NLR`	6	16319	82798	1364
13	`c6_opposite/LLL`	6	16363	81451	1518
16	`c6_opposite/LRL`	2	16527	80934	211
27	`c6_mixed/NNN`	1	15197	84067	388681
28	`c6_mixed/NLN`	2	15797	82439	33160
30	`c5_leaf_far/NNN`	1	14551	77874	7863
31	`c5_leaf_far/NNL`	2	14917	76098	1212
33	`c5_leaf_near/NNN`	1	14519	78032	5430
34	`square_tail/NNN`	1	15400	78367	1683
35	`square_adj_alternating/NNN`	1	14344	73287	1327
36	`square_adj_three/NNN`	1	14376	73415	1328
37	`square_opp_alternating/NNN`	1	13880	75565	1315
38	`square_opp_alternating/NNL`	2	14079	70981	1188
40	`square_opp_three/NNN`	1	13880	73652	5143
41	`k23_isolated/NNN`	1	14114	72740	1069
42	`triangle_tail/NNL`	2	16095	80145	1403
44	`two_triangles/NNL`	6	17507	88782	13124
47	`two_triangles/NLL`	6	18118	91283	5217
48	`two_triangles/NLR`	6	18334	93832	3281
56	`two_triangles/LLL`	6	18567	92141	7115
59	`two_triangles/LRL`	2	18939	91689	1178

D Companion proof concordance

The numbering in the numerical tables names the following exact mathematical sources. Their byte identities and individual premise statements are retained in the machine-readable source concordance. The separate generation proof and order-51 reduction are also included in that concordance.

ID	Companion mathematical proof
1	Cycle counts and explicit symmetry witnesses
2	Lower bounds from short cycles
3	Two minima among cubic graphs of girth nine
4	Minimum orders within two symmetry classes
5	Reconstruction from girth cycles
6	A small incidence classification for unique pentagons
7	Twenty extremal graphs of small order
8	Two small extremal graphs from the triangle-free degree bound

ID	Companion mathematical proof
9	Cubic graphs from complete catalogues
10	The graph of degree six with one triangle on each edge
11	A graph at the girth-six Moore bound
12	Four-cycle counts determine a fourteen-vertex graph
13	Common-neighbour counts and the Clebsch graph
14	A dense graph determined by its squares
15	Squares near a shortest odd cycle
16	Constant square counts on sixteen vertices
17	A Latin square in the bipartite complement
18	Excluding two odd orders by the same cycle partition
19	Square counts on twenty vertices
20	Eight squares through every edge
21	Distance layers and square moments
22	Square counts in graphs of degree five
23	Four squares through each edge
24	A graph from fourteen points and fourteen blocks
25	Three vertices outside a Moore tree
26	Three exterior vertices in a graph of girth five
27	Reusing partial graphs in a classification of order forty
28	Edges outside a Moore ball
29	Four vertices outside a Moore edge tree
30	Equality in the count of third neighbours
31	Integer third-layer counts
32	Eight pentagons through every edge
33	Seven pentagons through every edge
34	Fourteen octagons through every edge
35	Twenty-three hexagons through every edge
36	Twenty-two hexagons through every edge
37	Twenty hexagons through every edge: a finite lower bound
38	Nineteen hexagons through every edge: the remaining finite lower bound
39	Eighteen hexagons through every edge: the unique graph of order thirty-two
40	Fifteen hexagons through every edge: exclusion of order thirty-two
41	Thirteen hexagons through every edge: exclusion of order thirty-three
42	Fourteen hexagons through every edge: one exterior edge
43	Twelve hexagons through every edge: one exterior edge
44	Ten hexagons through every edge: three exterior edges
45	Quartic graphs with five pentagons through every edge
46	Quartic graphs with few pentagons through each edge
47	A lower bound for pentavalent graphs with fourteen pentagons on each edge
48	Two pentagons through every edge of a pentavalent graph
49	Cubic graphs with two octagons through every edge
50	Excluding order 54 with one hexagon through every edge

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[13] Human Compression project, *Cubic graphs from complete catalogues*. Companion component C9; exact source and acquired premises are listed in the source concordance.

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